

Summarization (CRQA : EEG)

Over the past decade, Cross-Recurrence Quantification Analysis (CRQA) and Recurrence Quantification Analysis (RQA) emerged as powerful nonlinear methods for processing EEG signals, because of the ability they offer on capturing complex dynamics in nonstationary data. These techniques have successfully helped in clinical and cognitive neuroscience applications by offering insights that traditional linear methods were unable to catch.

In epilepsy research, CRQA based approaches proposed new seizure detection biomarkers. The work of Rangaprakash[1], introduced the Correlation between Probabilities of Recurrence (CPR), a novel CRQA-derived phase synchronization metric which outperformed traditional linear coherence measures. CPR differentiates among interictal and preictal states while also can distinguish between eyes-open and eyes-closed conditions in epileptic patients. Building on this, Yang et al.[2], applied CRQA to stereoelectroencephalography (sEEG) recordings, revealing that epileptogenic channels present longer diagonal structures in their recurrence plots, indicating more deterministic and recurrent dynamics during pre-ictal periods. These findings, suggested that CRQA could identify pathological brain regions with greater accuracy than standalone spectral analysis.

For real-time seizure detection, Fan and Chou[3], have developed an new approach by blending recurrence plots with spectral graph theory. In their method authors tested the CHB-MIT dataset and achieved 98.48% sensitivity and very low detection latency (6 seconds). Gruszczyńska et al.[4] further demonstrated the clinical utilization of RQA by showing that standard RQA metrics (*determinism*, *laminarity*, and *longest diagonal line*) are able to classify epileptic EEGs with 86.8% accuracy even during interictal periods. This finding was particularly significant as it suggested that RQA could identify those pathological patterns without requiring actual seizure events during recording.

In the domain of cognitive decline and neurological diseases, CRQA/RQA methods also provided new biomarkers for early diagnosis or progression monitoring. Timothy et al.[5] in their study, combined RQA and CRQA in order to analyze both resting-state and memory-task EEG recordings in mild cognitive impairment (MCI). Findings of this work showed that MCI patients experienced lower complexity (quantified through RQA) and higher inter-hemispheric synchronization (measured via CRQA) compared to healthy controls, especially during cognitive tasks. Integration of these features achieved 91.7% classification accuracy, potentially opening the way for novel clinical diagnostic applications.

Nunez et al.[6], applied and extended these findings to Alzheimer's disease (AD), by employing a sophisticated approach by combined wavelet based analysis of nonstationarity with RQA metrics. Their results showed distinct recurrence patterns in theta and beta bands that differentiated AD patients from healthy controls. Notably, they found that entropy of recurrence point density and median recurrence point density served as particularly sensitive markers of neurodegeneration. Heunis et al.[7], applied similar methodology to study

autism spectrum disorder (ASD), finding that RQA features extracted from resting-state EEG can distinguish ASD children from typically developing controls with 92.9% accuracy using nonlinear SVM classifiers.

Considering cognitive neuroscience, CRQA/RQA has also helped with insights into brain dynamics during task performance. Frolov et al.[8], developed a multiplex network approach using CRQA to simultaneously analyze within-frequency and cross-frequency coupling in EEG during working memory tasks. Their method highlighted previously undetectable patterns in dynamic connectivity modulation which correlated with task difficulty and performance. Guglielmo et al.[9] focused on mental arithmetic tasks, extracting six key RQA metrics (including *recurrence rate*, *determinism*, and *laminarity*) from frontal-parietal electrodes. By application of a machine learning pipeline afterwards, they reached over 85% of accuracy in classifying cognitive performance levels, highlighting RQA’s potential for real time cognitive state monitoring.

Recent methodological innovations continue to expand the potential applications of these techniques. For instance, the integration of CRQA with machine learning (as demonstrated by [9] and [10]) has opened new possibilities for automated diagnosis and real time monitoring systems. On top of that, the development of network-based approaches ([8] and [11]) has extended recurrence analysis from single-channel characterization to comprehensive assessments of whole-brain dynamics.

The described works, establish CRQA and RQA as novel tools on the domain of EEG analysis, with applications diversing from epilepsy monitoring to cognitive assessment and neurodegenerative disease diagnosis. The consistent useful results in clinical and experimental paradigms suggests these methods can play an important role for research and clinical practice as the field moves into new sophisticated analysis of brain dynamics.

References

- [1] Rangaprakash, D. (2014). Connectivity analysis of multichannel EEG signals using recurrence based phase synchronization technique. *Computers in Biology and Medicine*, 46, 11-21. <https://doi.org/10.1016/j.compbiomed.2013.10.025>
- [2] Yang, C., Luan, G., Liu, Z. and Wang, Q. (2019). Dynamical analysis of epileptic characteristics based on recurrence quantification of SEEG recordings. *Physica A: Statistical Mechanics and its Applications*, 523, 507–515. <https://doi.org/10.1016/j.physa.2019.02.017>
- [3] Fan, M. and Chou, C.-A. (2019). Detecting abnormal pattern of epileptic seizures via temporal synchronization of EEG signals. *IEEE Transactions on Biomedical Engineering*, 66(3), 601-608. <https://doi.org/10.1109/TBME.2018.2850959>
- [4] Gruszczyńska, I., Mosdorf, R., Sobaniec, P., Żochowska-Sobaniec, M. and Borowska, M. (2019). Epilepsy identification based on EEG signal using RQA method. *Advances in Medical Sciences*, 64(1), 58-64. <https://doi.org/10.1016/j.advms.2018.08.003>
- [5] Timothy, L. T., Krishna, B. M. and Nair, U. (2017). Classification of mild cognitive impairment EEG using combined recurrence and cross recurrence quantification analysis. *International Journal of Psychophysiology*, 120, 86–95. <https://doi.org/10.1016/j.ijpsycho.2017.07.006>
- [6] P. Núñez, J. Poza, C. Gómez, V. Barroso-García, A. Maturana-Candelas, M. A. Tola-Arribas, M. Cano, and R. Hornero, Characterization of the dynamic behavior of neural activity in Alzheimer’s disease: exploring the non-stationarity and recurrence structure of EEG resting-state activity, *Journal of Neural Engineering*, vol. 17, no. 1, p. 016071, 2020, doi: <https://doi.org/10.1088/1741-2552/ab71e9>.
- [7] Heunis, T., Aldrich, C., Peters, J. M., Jeste, S. S., Sahin, M., Scheffer, C., and de Vries, P. J., “Recurrence quantification analysis of resting state EEG signals in autism spectrum disorder – a systematic methodological exploration of technical and demographic confounders in the search for biomarkers,” *BMC Medicine*, vol. 16, no. 1, p. 101, 2018.
- [8] Frolov, N., Maksimenko, V. and Hramov, A. (2020), *Revealing a multiplex brain network through the analysis of recurrences* Chaos: An Interdisciplinary Journal of Nonlinear Science 30(12), 121108.
- [9] Guglielmo, G., Wiltshire, T. J. and Louwerse, M. (2022). Training machine learning models to detect group differences in neurophysiological data using recurrence quantification analysis based features. *ICAART 2022 - 14th International Conference on Agents and Artificial Intelligence*, 430-434.

- [10] Vanithamani Palanisamy, A. Ranichitra, and Radhamani Ellapparaj, A differential biomarker based on recurrence quantification analysis of EEG signal and genetic algorithm for epilepsy diagnosis. *Journal of Artificial Intelligence and System Modelling*, 2(2):74–85, 2024. 10.22034/jaism.2024.450400.1046
- [11] Lopes, M. A., Zhang, J., Krzeminski, D., Hamandi, K., Chen, Q., Livi, L. and Masuda, N. (2020). Recurrence quantification analysis of dynamic brain networks. *European Journal of Neuroscience*, 00, 1-20. <https://doi.org/10.1111/ejn.14960>